Solution to Challenge 7

A Limit Cycle in the Van der Pol System

(Alligood, Sauer & Yorke, *Chaos: An Introduction to Dynamical Systems*, Ch. 7)

Overview. The Van der Pol equation $\ddot{x} + (x^2 - 1)\dot{x} + x = 0$ is turned into a system whose phase plane has a clean "circulation" structure (Steps 1–2). Following one revolution around the origin defines a first-return map T on the positive z-axis (Step 3). A periodic orbit is the same as a fixed point of T. We find the fixed point by tracking the change in the energy $E = \frac{1}{2}(x^2 + z^2)$ over half a revolution (Steps 4–10), and then show that it attracts (Step 11).

Throughout, "uniqueness" means the Existence and Uniqueness Theorem for ODEs (all vector fields here are polynomial, hence smooth), and a dot is d/dt.

Step 1: The change of coordinates

Let $F(x) = \dfrac{x^3}{3} - x$ and $z = y + F(x)$.

One-to-one. The map $\Psi(x, y) = (x,\, y + F(x))$ has the explicit inverse $\Psi^{-1}(x, z) = (x,\, z - F(x))$. Both are smooth, so Ψ is a smooth bijection of $\mathbb{R}^2$ with smooth inverse: a legitimate change of coordinates.

The transformed system. From (7.50), $\dot{x} = y = z - F(x)$. Also

$$\dot{z} = \dot{y} + F'(x)\dot{x} = \big[-x + (1 - x^2)y\big] + (x^2 - 1)y = -x.$$

Hence

$$\boxed{\dot{x} = z - \left(\tfrac{x^3}{3} - x\right) = z - F(x), \qquad \dot{z} = -x} \tag{1}$$

which is (7.51).

Two facts used repeatedly.

(a) *Equilibria.* (7.51) has $\dot{z} = 0 \Rightarrow x = 0$ and then $\dot{x} = 0 \Rightarrow z = 0$: the origin is the only equilibrium. (For (7.50) the linearization at 0 is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$, trace 1, determinant 1, eigenvalues $\frac{1}{2}(1 \pm i\sqrt{3})$: an unstable spiral.)

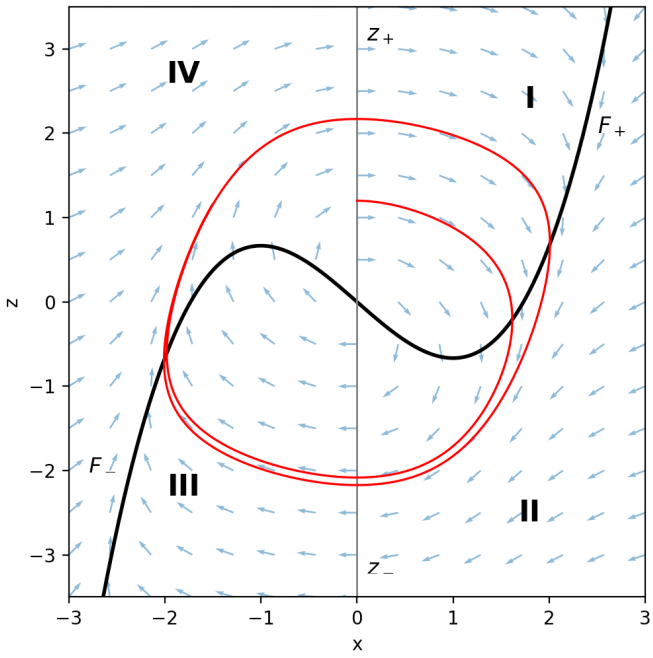
(b) *Symmetry.* F is odd. Let $\rho(x, z) = (-x, -z)$. If $(x(t), z(t))$ solves (7.51) then so does $(-x(t), -z(t))$: $\frac{d}{dt}(-x) = -(z - F(x)) = (-z) - F(-x)$ and $\frac{d}{dt}(-z) = x = -(-x)$. So ρ maps orbits to orbits.

Also $F(x) \geq -\frac{2}{3}$ for $x \geq 0$ (minimum at $x = 1$), $F < 0$ on $(0, \sqrt{3})$, $F(\sqrt{3}) = 0$, and F is strictly increasing on $[1, \infty)$ (since $F' = x^2 - 1$).

Step 2: The phase plane and the circulation

Nullclines and regions. $\dot{x} = 0$ exactly on the curve $z = F(x)$ (F_+ for $x > 0$, F_- for $x < 0$) and $\dot{z} = 0$ exactly on the z-axis $x = 0$. These split the plane into four regions:

$$\begin{array}{lllll} \text{I}: & x > 0,\ z > F(x) & \dot{x} > 0,\ \dot{z} < 0 & (\rightarrow,\ \downarrow) \\ \text{II}: & x > 0,\ z < F(x) & \dot{x} < 0,\ \dot{z} < 0 & (\leftarrow,\ \downarrow) \\ \text{III}: & x < 0,\ z < F(x) & \dot{x} < 0,\ \dot{z} > 0 & (\leftarrow,\ \uparrow) \\ \text{IV}: & x < 0,\ z > F(x) & \dot{x} > 0,\ \dot{z} > 0 & (\rightarrow,\ \uparrow) \end{array}$$



Regions I–IV, the curves $F_\pm$, the half-axes $z_\pm$, the direction field of (7.51), and (red) one solution starting on z_+.

Crossing directions on the boundaries.

- On $z_+ = \{x = 0, z > 0\}$: $\dot{x} = z > 0$, so the flow passes from IV into I.
- On $z_- = \{x = 0, z < 0\}$: $\dot{x} = z < 0$, so the flow passes from II into III.
- With $g = z - F(x)$ one computes $\dot{g} = \dot{z} - F'(x)\dot{x} = -x - (x^2 - 1)g$. On F_+ ($g = 0$, $x > 0$) $\dot{g} = -x < 0$: the flow passes from I into II. On F_-, $\dot{g} = -x > 0$: from III into IV.

So the flow can only circulate I $\rightarrow$ II $\rightarrow$ III $\rightarrow$ IV $\rightarrow$ I (clockwise).

Claim. A solution starting at $(0, a^+)$, $a^+ > 0$, passes through I, crosses F_+, passes through II, crosses z_-, passes through III, crosses F_-, passes through IV, and returns to z_+; then the pattern repeats.

Region I is left in finite time through F_+. Start at $(0, a)$, $a > 0$. Since $\dot{x}(0) = a > 0$ the solution enters I. While it is in I, x is increasing and z is decreasing, and $z > F(x) \geq -\frac{2}{3}$. Moreover $\dot{x} = z - F(x) \leq a + \frac{2}{3}$, so x cannot blow up in finite time; the solution exists as long as it stays in I. Suppose it stayed in I for all $t \geq 0$. Fix $t_0 > 0$ and let $\varepsilon = x(t_0) > 0$. Then for $t \geq t_0$, $\dot{z} = -x \leq -\varepsilon$, so $z(t) \rightarrow -\infty$, contradicting $z > -\frac{2}{3}$. Hence the solution leaves I at a finite time. It cannot exit through the z-axis (there $x \geq \varepsilon > 0$) nor at the origin (an equilibrium is never reached in finite time), so it exits through a point $(r, F(r)) \in F_+$ with $r > 0$. Call $r = r_a$.

Region II is left in finite time through z_-. Enter II at $(r, F(r))$. Since the flow cannot cross F_+ from II back into I, the solution can leave II only through z_-. Suppose it stays in II for all later times. Then x is decreasing and positive (so $x \in (0, r]$) and z is decreasing.

- If z is bounded below, (x, z) converges, and the limit of a solution is an equilibrium, i.e. the origin. Then eventually $x < \sqrt{3}$ and $z < F(x) < 0$ with z decreasing, so z cannot converge to 0. Contradiction.
- If $z \to -\infty$, then $\dot{x} = z - F(x) \to -\infty$ (F is bounded on $(0, r]$), so x reaches 0 in finite time. Contradiction.

So the solution reaches a point $(0, a^-)$ with $a^- < 0$ ($a^- \neq 0$, since the origin is an equilibrium).

Regions III and IV. By the symmetry ρ, the solution from $(0, a^-)$ is ρ applied to the solution from $(0, -a^-) \in z_+$. The latter goes through I and II and hits z_- at some $(0, a')$. Hence the former goes through III (its image of I), crosses F_-, goes through IV, and hits z_+ at $(0, -a')$. ■

The maps S and T. For $a > 0$ write the first hit of z_- as $(0, -S(a))$, so $S(a) > 0$ is the "half-return map" $z_+ \to z_-$ (read as a positive number). The computation above shows

$$T(a) = S\big(S(a)\big), \tag{2}$$

because the solution from $(0, -S(a))$ is the ρ-image of the solution from $(0, S(a))$, which hits z_- at $(0, -S(S(a)))$, hence the first return to z_+ is at $(0, S(S(a)))$. The solution meets z_+ only when passing from IV to I, so $T(a)$ is indeed the first return after one full cycle.

Step 3: T is one-to-one; fixed points ⇔ periodic orbits

A lemma: orbits through z_+ are nested

For $a > 0$ the arc $X_0(a)$ from $(0, a)$ to $(0, -S(a))$ lies in $x \geq 0$, and along it $\dot{z} = -x < 0$ except at the endpoints, so z strictly decreases. Therefore $X_0(a)$ is the graph of a function

$$x = \xi_a(z), \qquad -S(a) < z < a, \qquad \xi_a > 0, \;\; \xi_a(z) \to 0 \text{ at both ends},$$

and since $dx/dz = \dot{x}/\dot{z}$,

$$\frac{dx}{dz} = \frac{F(x) - z}{x} \qquad (x > 0). \tag{3}$$

The right side is smooth for $x > 0$, so two solutions of (3) either coincide or their graphs never touch. Likewise, using x as the variable on the two parts of the arc, the part in I is a graph $z = \varphi_a(x)$ and the part in II is a graph $z = \psi_a(x)$, $0 \leq x \leq r_a$, where

$$\varphi_a(0) = a, \;\; \varphi_a > F \text{ on } [0, r_a), \;\; \varphi_a(r_a) = \psi_a(r_a) = F(r_a), \;\; \psi_a < F \text{ on } [0, r_a), \;\; \psi_a(0) = -S(a),$$

and both solve

$$\frac{dz}{dx} = \frac{-x}{z - F(x)} \qquad (z \neq F(x)), \tag{4}$$

again with a smooth right side on each of $\{z > F\}$, $\{z < F\}$. Note that φ_a is decreasing in x (along I, x increases while z decreases) and ψ_a is increasing in x (along II, x and z both decrease).

Lemma 1 (nesting)**.** Let $0 < a < b$. Then

(i) $\xi_a < \xi_b$ on $(-S(a), a)$ and $S(a) < S(b)$;
(ii) $r_a < r_b$;
(iii) $\varphi_a < \varphi_b$ on $[0, r_a]$ and $\psi_a > \psi_b$ on $[0, r_a]$.

In words: the orbit through the higher starting point lies strictly outside, at every height and at every x.

Proof. (i) Near $z = a$ we have $\xi_a(z) \to 0$ while $\xi_b(a) > 0$ (note $a \in (-S(b), b)$), so $\xi_a < \xi_b$ for z slightly below a. If $\xi_a = \xi_b$ at some z of the common interval $J = (\max(-S(a), -S(b)), a)$, uniqueness for (3) would force them to agree on all of J, contradicting the inequality near a. So $\xi_a < \xi_b$ on J.

Suppose $S(b) < S(a)$. Then $J = (-S(b), a)$ and $z_0 = -S(b)$ lies inside $(-S(a), a)$ so $\xi_a(z_0) > 0$. Letting $z \downarrow z_0$ in $\xi_a < \xi_b$ gives $\xi_a(z_0) \le \lim \xi_b = 0$, a contradiction. Suppose $S(b) = S(a) =: s$. Then both solutions pass through $(0, -s)$ and are therefore the same orbit. Going backward in time from $(0, -s)$, the first point on z_+ is unique, and both $(0, a)$ and $(0, b)$ would be that point (neither arc X_0 contains another point of z_+), so $a = b$, a contradiction. Hence $S(a) < S(b)$ and $J = (-S(a), a)$.

(ii) Let z^* be the height where ξ_a attains its maximum r_a (an interior point). Then $r_a = \xi_a(z^*) < \xi_b(z^*) \le r_b$.

(iii) Let $m = r_a < r_b$. On $[0, m)$, φ_a, φ_b solve (4) in $\{z > F\}$ and $\varphi_a(0) = a < b = \varphi_b(0)$. By uniqueness they never meet, so $\varphi_a < \varphi_b$ on $[0, m)$, and by continuity $\le$ at m (in fact $<$, since $\varphi_b(m) > F(m) = \varphi_a(m)$ as $m < r_b$). Similarly ψ_a, ψ_b solve (4) in $\{z < F\}$ on $[0, m)$ and $\psi_a(0) = -S(a) > -S(b) = \psi_b(0)$ by (i), so $\psi_a > \psi_b$ on $[0, m)$; at m, $\psi_a(m) = F(m) > \psi_b(m)$. □

Consequences

- **S and T are strictly increasing, so $T = S \circ S$ is one-to-one.** (Directly: if $T(a) = T(b)$ with $a \neq b$, Lemma 1 gives $T(a) \neq T(b)$.)
- **$T(a) = a \implies$ periodic.** If $T(a) = a$, the solution through $(0, a)$ is back at $(0, a)$ after a positive time τ; since (7.51) is autonomous, $v(t + \tau) = v(t)$: it is periodic.
- **Periodic $\implies T(a) = a$.** Let v be a periodic solution through $(0, a)$ with $v(\tau) = v(0)$, $\tau > 0$. The solution meets z_+ exactly at the successive returns $(0, T^n(a))$, $n \ge 1$, so $v(\tau) = (0, a)$ means $T^n(a) = a$ for some $n \ge 1$. As T is strictly increasing, $T(a) > a$ would give $T^n(a) > a$ and $T(a) < a$ would give $T^n(a) < a$. So $T(a) = a$. (Every periodic orbit crosses z_+, by the Claim of Step 2, since every non-equilibrium orbit cycles through I–IV.)
- **(7.50) versus (7.51).** Ψ is a bijection and carries solutions of (7.50) to solutions of (7.51) (Step 1). So $v(t)$ is a solution of (7.50) with $v(\tau) = v(0)$ iff $\Psi(v(t))$ is a solution of (7.51) with the same property: periodic orbits correspond, with the same period.

Continuity. S and T are continuous: the crossing of z_- is transversal ($\dot{x} = z \neq 0$ there), so the crossing time and point depend continuously on the initial condition by Continuous Dependence (Theorem 7.16) and the Implicit Function Theorem. This will be needed in Step 10.

Step 4: Symmetry gives a fixed point

Suppose the solution through $(0, a^+)$, $a^+ > 0$, hits z_- at $(0, -a^+)$, i.e. $S(a^+) = a^+$. Then by (2), $T(a^+) = S(S(a^+)) = S(a^+) = a^+$. Geometrically: the solution from $(0, -a^+)$ is ρ of the solution from $(0, a^+)$, so it retraces the first half reflected through the origin and lands at $(0, a^+)$.

Define the energy $E(x, z) = \frac{1}{2}(x^2 + z^2)$ and

$$h(a^+) = E(0, a^-) - E(0, a^+) = \tfrac{1}{2}\big[(a^-)^2 - (a^+)^2\big] = \tfrac{1}{2}\big[S(a^+)^2 - (a^+)^2\big],$$

where $a^- = -S(a^+)$. Because $S > 0$,

$$h(a) > 0 \iff S(a) > a, \qquad h(a) = 0 \iff S(a) = a, \qquad h(a) < 0 \iff S(a) < a. \tag{5}$$

So it suffices to find $c > 0$ with $h(c) = 0$.

Step 5: The energy identity

Along solutions of (7.51),

$$\dot{E} = x\dot{x} + z\dot{z} = x\big(z - F(x)\big) - zx = -xF(x) = F(x)\,\dot{z} = \Big(\tfrac{x^3}{3} - x\Big)\dot{z}.$$

This is the claimed formula. Equivalently

$$\dot{E} = -xF(x) = x^2\Big(1 - \frac{x^2}{3}\Big) \quad \Longrightarrow \quad \dot{E} > 0 \text{ for } 0 < |x| < \sqrt{3}, \qquad \dot{E} < 0 \text{ for } |x| > \sqrt{3}. \tag{6}$$

Therefore

$$h(a^+) = \int_{X_0} \dot{E}\,dt = \int_{X_0} \Big(\frac{x^3}{3} - x\Big)\dot{z}\,dt.$$

Let $q = \sqrt{3}$, the positive zero of F (so F_+ meets the x-axis at $(q, 0)$), and recall $r = r_{a^+}$ is the x-coordinate where X_0 meets F_+.

Step 6: If $r < q$ then $h > 0$

Along X_0, which lies in $x \geq 0$ with x increasing in I up to r and decreasing in II after r, we have $0 \leq x \leq r < q = \sqrt{3}$. By (6), $\dot{E} = x^2(1 - x^2/3) > 0$ except at the two endpoints where $x = 0$. Hence $h(a^+) = \int_{X_0} \dot{E}\,dt > 0$. (The same holds if $r = q$: then $x \leq q$ with equality at one instant only.)

Step 7: If $r > q$, split X_0 into three pieces

Let $r > q$. The line $x = q$ is met twice by X_0: first at q^+ in region I (on the way out), then at q^- in region II (on the way back). Write $X_0 = X_1 \cup X_2 \cup X_3$ with

$$X_1 : (0, a^+) \to q^+, \qquad X_2 : q^+ \to q^-, \qquad X_3 : q^- \to (0, a^-).$$

Using $x \uparrow$ in I and $x \downarrow$ in II:

- on X_1, $0 \leq x \leq q$, so $\dot{E} > 0$ (apart from the endpoint $x = 0$): $\displaystyle\int_{X_1} dE > 0$;
- on X_2, $x \geq q$ (it goes up to $x = r$ and back down), so $\dot{E} < 0$ except at the endpoints: $\displaystyle\int_{X_2} dE < 0$;
- on X_3, $0 \leq x \leq q$: $\displaystyle\int_{X_3} dE > 0$.

Thus $h = h_1 + h_2 + h_3$ with $h_1, h_3 > 0 > h_2$, where $h_i(a^+) = \int_{X_i} \dot{E}\,dt$.

Step 8: h_1 and h_3 are decreasing in a^+

Let $a < b$ both have $r > q$.

X_1. On X_1, x is a valid parameter. With $\dot{x} = z - F(x)$ we have $dt = dx/(z - F(x))$ and so

$$h_1(a) = \int_{X_1} F(x)\,\dot{z}\,dt = \int_{X_1} F(x)\,\frac{-x}{z - F(x)}\,dx = \int_0^q \frac{x^2\big(1 - \frac{x^2}{3}\big)}{\varphi_a(x) - F(x)}\,dx.$$

(This is the hint's computation.) The numerator is positive on $(0, q)$. By Lemma 1(iii), $\varphi_a < \varphi_b$ on $[0, q]$, so the denominators for b are larger and

$$h_1(b) < h_1(a).$$

X_3. On X_3 the solution lies in II, $\dot{x} < 0$, $dt = dx/(\psi - F)$ with $\psi - F < 0$, and x runs from q down to 0:

$$h_3(a) = \int_{x=q}^0 \frac{x^2(1 - \frac{x^2}{3})}{\psi_a(x) - F(x)}\,dx = \int_0^q \frac{x^2\big(1 - \frac{x^2}{3}\big)}{F(x) - \psi_a(x)}\,dx.$$

By Lemma 1(iii), $\psi_a > \psi_b$, so $F - \psi_b > F - \psi_a > 0$ and again

$$h_3(b) < h_3(a).$$

So both are strictly decreasing functions of a^+ (on the set where $r > q$).

Step 9: h_2 decreases monotonically to $-\infty$

On X_2 (which lies in $x > 0$) z is strictly decreasing, so X_2 is the part of the graph $x = \xi_a(z)$ with $z \in [z_2(a), z_1(a)]$, where $z_1(a) = \varphi_a(q)$ and $z_2(a) = \psi_a(q)$ are the heights of q^+, q^-. Using $\dot{E}\,dt = F(x)\,dz$,

$$h_2(a) = \int_{X_2} F(x)\,dz = \int_{z_1}^{z_2} F(\xi_a(z))\,dz = -\int_{z_2(a)}^{z_1(a)} F(\xi_a(z))\,dz,$$

and on this range $\xi_a \geq q$, so $F(\xi_a) \geq 0$ and $h_2 \leq 0$. (This is the hint: $h_2 = -\int_{X_2}(\frac{x^3}{3} - x)\,x\,dt$.)

Monotone. Let $a < b$ with $r_a > q$. By Lemma 1(iii), $z_1(a) = \varphi_a(q) < \varphi_b(q) = z_1(b)$ and $z_2(a) = \psi_a(q) > \psi_b(q) = z_2(b)$: the z-interval for b strictly contains that for a. On the smaller interval, $q \leq \xi_a < \xi_b$ by Lemma 1(i), and F is strictly increasing on $[1, \infty)$, so $F(\xi_a) < F(\xi_b)$; on the extra parts $F(\xi_b) \geq 0$. Hence $\int F(\xi_b)\,dz > \int F(\xi_a)\,dz$, i.e. $h_2(b) < h_2(a)$.

Unbounded orbits. First, $r_a \to \infty$ as $a \to \infty$. Fix $R > 0$ and let $M = \max_{[0,R]} |F|$. If $r_a \leq R$ then, while in I, $\dot{z} = -x \geq -R$, so $z(t) \geq a - Rt$, and for $a \geq M + 2R$ the solution is still in I at $t = 1$ (it can only exit at $z = F(r_a) \leq M$, but $z(1) \geq a - R > M$). Then $x(1) = \int_0^1 (z - F(x))\,dt \geq \int_0^1 (a - Rt - M)\,dt = a - M - \frac{R}{2} > R$, contradicting $x \leq r_a \leq R$. So $r_a > R$ for all large a.

Now fix $x_0 = 2 > q$ (so $F(x_0) = \frac{2}{3}$, and $F \leq \frac{2}{3}$ on $[0, 2]$). Choose a_0 with $r_{a_0} > 2$, and set $m' = \varphi_{a_0}(2) - F(2) > 0$. For $a \geq a_0$ and $x \in [0, 2]$, $\varphi_a(x) \geq \varphi_a(2) \geq \varphi_{a_0}(2)$, so $\varphi_a(x) - F(x) \geq m'$ and from (4)

$$a - \varphi_a(2) = \int_0^2 \frac{x}{\varphi_a(x) - F(x)}\,dx \leq \frac{2}{m'}, \qquad \text{i.e.} \qquad \varphi_a(2) \geq a - \frac{2}{m'}.$$

The part of X_2 with $x \geq 2$ covers the heights $z \in [\psi_a(2), \varphi_a(2)]$, and there $F(\xi_a(z)) \geq F(2) = \frac{2}{3}$. Also $\psi_a(2) < F(2) = \frac{2}{3}$. Therefore

$$-h_2(a) \;\geq\; \tfrac{2}{3}\big(\varphi_a(2) - \psi_a(2)\big) \;\geq\; \tfrac{2}{3}\Big(a - \tfrac{2}{m'} - \tfrac{2}{3}\Big) \;\longrightarrow\; \infty,$$

so $h_2(a) \to -\infty$ as $a \to \infty$.

Step 10: A unique fixed point c

Let $U = \{a > 0 :\ r_a > q\}$. By Lemma 1(ii), U is an interval $\{a > a_*\}$ or $\{a \geq a_*\}$, and by Step 9, $U \neq \emptyset$.

Small a lie outside U. (So $h > 0$ near 0.) Scale $x = aX$, $z = aZ$. Then $\dot{X} = Z + X - \frac{a^2}{3}X^3$, $\dot{Z} = -X$, $(X, Z)(0) = (0, 1)$, which tends as $a \to 0$ to the linear system $\dot{X} = X + Z$, $\dot{Z} = -X$ (eigenvalues $\frac{1}{2}(1 \pm i\sqrt{3})$). Its solution $X(t) = \frac{2}{\sqrt{3}}e^{t/2}\sin\frac{\sqrt{3}}{2}t$ reaches $\dot{X} = 0$ (the analogue of crossing F_+, which scales to $Z = -X + \frac{a^2}{3}X^3 \to -X$) at $t_1 = \frac{2\pi}{3\sqrt{3}}$, transversally ($\ddot{X} = -X < 0$). By continuous dependence, the actual crossing of F_+ happens at a bounded scaled amplitude, so $r_a = O(a) \to 0$. Hence $r_a < q$ for small a.

Conclusion.

- For $a \notin U$ ($r_a \leq q$): $h(a) > 0$ by Step 6. Such a exist (small a).
- For $a \in U$: $h = h_1 + h_2 + h_3$ is strictly decreasing by Steps 8–9, and $h(a) \to -\infty$ as $a \to \infty$ (since $h_1, h_3 \leq h_1(a_0), h_3(a_0)$ stay bounded above while $h_2 \to -\infty$).
- h is continuous (it is $\frac{1}{2}[S(a)^2 - a^2]$ and S is continuous).

By the Intermediate Value Theorem h has a zero c, necessarily in U (off U it is positive). It is unique, because h is strictly decreasing on U. By (5), $S(c) = c$, i.e. $a^- = -c$, and by Step 4, $T(c) = c$. By Step 3 the solution through $(0, c)$ is a periodic orbit Γ of (7.51) (and, via Ψ^{-1}, of (7.50)), and it is the *only* one: any periodic orbit meets z_+ at a fixed point of T.

Numerical check. Integrating (7.51) gives $c \approx 2.1727$; in the original variables the limit cycle crosses the y-axis at $(0, \pm c)$ (as $F(0) = 0$, $z = y$ there) and its maximal x is about 2.0.

Step 11: c is attracting

By (5) and Step 10,

$$S(a) > a \quad (0 < a < c), \qquad S(a) < a \quad (a > c).$$

Uniqueness gives nesting. Let $0 < a < c$. Since S is strictly increasing (Lemma 1) and $S(c) = c$,

$$S(a) < S(c) = c,$$

i.e. $-a^- < c$ (the solution from $(0, a)$ crosses z_- strictly inside $(0, -c)$; geometrically it stays inside the cycle Γ). Now $a < S(a) < c$. Apply the same to $a' = S(a) \in (0, c)$:

$$S(a) < S(S(a)) = T(a) < c.$$

Hence

$$0 < a < S(a) < T(a) < c \qquad (0 < a < c).$$

In the same way, for $a > c$ we get $c < T(a) < S(a) < a$.

Energy. Over one full turn the energy changes by

$$E(T(a)) - E(a) = \tfrac{1}{2}\big[T(a)^2 - a^2\big] \begin{cases} > 0 & 0 < a < c, \\ < 0 & a > c, \end{cases}$$

which is the hint's statement: energy increases for orbits starting inside the cycle and decreases for those starting outside.

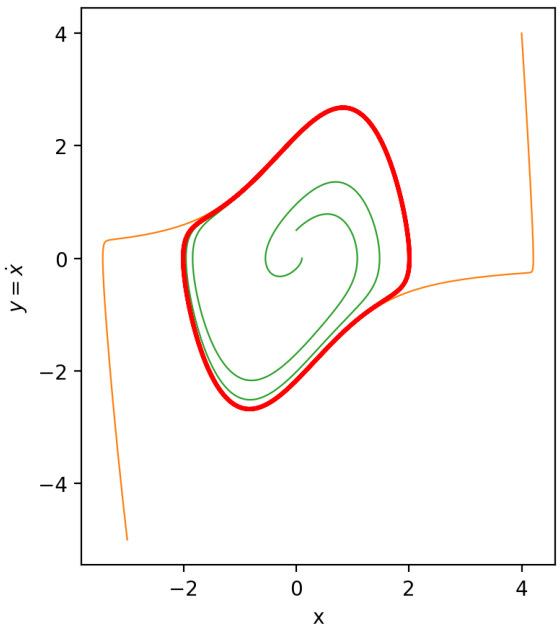
Convergence. Let $0 < a < c$. Then $a_n = T^n(a)$ is strictly increasing (since $T(a') > a'$ for every $a' \in (0, c)$) and bounded by c, so it converges to some $L \le c$. By continuity of T, $T(L) = L$. The only fixed point of T in $(0, \infty)$ is c (if $T(L) = L$ with $S(L) > L$ then $T(L) = S(S(L)) > S(L) > L$; if $S(L) < L$ then $T(L) < S(L) < L$; so $S(L) = L$ and $L = c$ by Step 10). So $T^n(a) \to c$. For $a > c$ the sequence decreases to c in the same way. Hence

$$\boxed{T^n(a) \to c \quad \text{for every } a > 0,}$$

and c is an attracting fixed point of T (with basin the whole half-axis).

Conclusion: the limit cycle

- Every solution of (7.51) other than the origin cycles around the origin through I–IV (Step 2 applies to any starting point not at the origin) and crosses z_+ at points $T^n(a) \to c$.
- Between successive crossings the solution follows a continuous image of one revolution, so by continuous dependence on initial conditions over one period it comes closer and closer to Γ.
- Thus every solution of the Van der Pol system (7.50), except the unstable equilibrium at the origin, approaches the unique periodic orbit $\Psi^{-1}(\Gamma)$ as $t \to \infty$: a limit cycle.



Van der Pol orbits in the original (x, y) plane: orbits starting inside (green) and outside (orange) the limit cycle (red) are attracted to it.

Why this is a nonlinear phenomenon. For a linear system with a periodic orbit (e.g. $\ddot{x} + x = 0$) the nearby orbits are also periodic, so no orbit is attracted to a single cycle; here the damping $(x^2 - 1)$ is *negative* for small $|x|$ (energy is pumped in: $\dot{E} > 0$ for $|x| < \sqrt{3}$) and *positive* for large $|x|$ (energy is removed: $\dot{E} < 0$ for $|x| > \sqrt{3}$), and the balance between the two occurs on exactly one cycle.